EMPIRICAL COMPARATIVE ANALYSIS

PyTorch vs. TensorFlow: Architectural Parity & Runtime Realities

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Slide 1: Core Design Philosophies

- **PyTorch (Define-by-Run):** Dynamic execution graph built on-the-fly. Enables Pythonic debugging and rapid academic velocity.
- **TensorFlow (Define-and-Run):** Static computation graph optimized for scale. Serves as the industry standard for robust enterprise deployment.

Slide 2: Experimental Methodology ($N = 10$ Runs)

- **Dataset:** MNIST 28x28 grayscale images.
- **Architecture:** Conv2D (32 filters, 3x3), MaxPool, Flatten, Dense (10 classes).
- **Hyperparameters:** Adam optimizer ($lr = 0.001$), batch size 64, trained for 3 epochs.
- **JIT Compilers:** `torch.compile()` vs. TensorFlow XLA.

Slide 3: Training Runtime Performance

- **PyTorch (`torch.compile`): 109.69s** mean training time ($\pm$2.86s std dev).
- **TensorFlow (XLA): 185.23s** mean training time ($\pm$12.60s std dev).

Slide 4: Predictive Accuracy Parity

- **PyTorch Accuracy: 98.05%** test set mean ($\pm$0.11% dev).
- **TensorFlow Accuracy: 97.98%** test set mean ($\pm$0.16% dev).

Slide 5: JIT Compilation Mechanics

- **TorchDynamo & Caching:** Dynamic graph evaluation with seamless fallback. Full graph support achieved without runtime interruption.
- **TensorFlow XLA Decorator:** Strict static graph optimization at the C++ level. Higher initialization overhead across benchmark runs.

Slide 6: Developer Ergonomics & Integration

- **PyTorch OOP Control:** Explicit module definition (`nn.Module`) and manual device allocation. Optimal velocity for algorithmic research and edge devices.
- **TensorFlow Keras Abstraction:** High-level sequential APIs abstracting layer construction. Enterprise microservices and cloud scaling standard.

Slide 7: Strategic Takeaway & Conclusion

- **No Universal Superiority:** Context Dictates Choice.
- PyTorch excels in experimental velocity and consistent compilation runtime.
- TensorFlow provides robust production pipelines for enterprise deployment.